DEFAKE : DETECTING GENAI & SOPHISTICATED MANIPULATION IN PUBLIC MEDIA

TRACK A: SOCIAL MEDIA & INFLUENCER AUTHENTICITY

TEAM MEMBERS

Yara Yaseer Alzahrani lamyaraalzahrani@gmail.com

Shouq Tareq Alhatem sho17ooq@gmail.com

Manar Eyad Ali manareyad689@gmail.com

PROBLEM STATEMENT

Recent advances in generative and facial manipulation methods have significantly reduced the effectiveness of conventional authenticity detection systems. While pretrained deepfake classifiers can assign images to AI-generated, Deepfake, or Real classes, their decisions are often driven by dataset-specific artifacts and lack interpretability, resulting in poor generalization to high-quality or unseen manipulations.

PROPOSED SOLUTION

In **Figure 1**, We introduce a hybrid detection architecture that augments a pretrained deepfake classification model with explicit forensic signal extraction, including **Local Phase Patterns (LPP), Laplacian residuals, and frequency-domain features obtained via FFT and DCT.** These low-level signals, together with the classifier's output probabilities, are provided to a Vision-Language Model (VLM) that performs semantic and physical consistency reasoning to validate and explain the predicted manipulation type.

METHODOLOGY

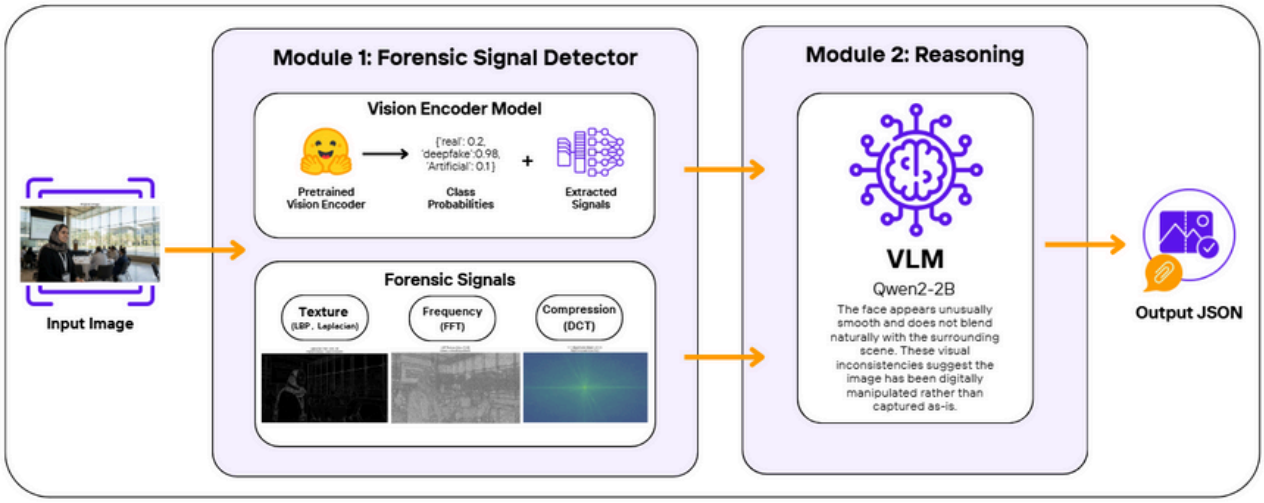


Figure 1: Dual-module authenticity detection architecture in which a pretrained vision encoder and explicit forensic signals (texture, frequency, and compression artifacts) are combined and provided to a Vision-Language Model for reasoning and generation of interpretable authenticity predictions.

Module 1: Forensic Signal Detector

Module 1 extracts low-level forensic evidence indicative of image manipulation or synthetic generation. An input image is processed by the pretrained vision encoder **prithivMLmods/AI-vs-Deepfake-vs-Real-9999**, which outputs class probabilities for **AI-generated**, **Deepfake**, and **Real** categories. In parallel, handcrafted forensic signals are extracted to capture statistical artifacts that may not be reliably learned by neural networks. These include texture-based features using **Local Phase Patterns (LPP)** and **Laplacian** filters, frequency-domain features obtained via **FFT**, and compression-related artifacts derived from **DCT** analysis. Together, the classifier outputs and forensic signals form a pixel-level representation of potential manipulation, as shown in **Figure 2**.

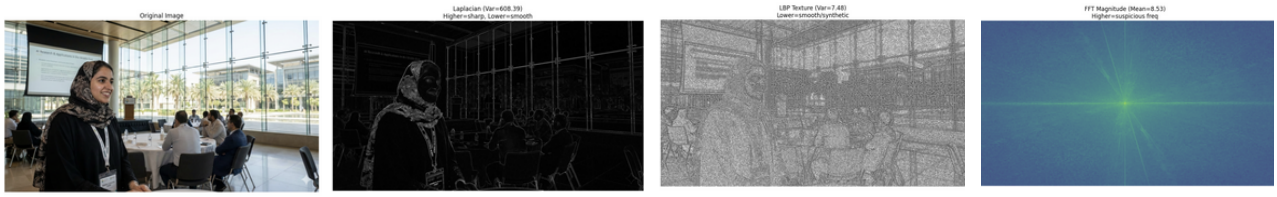


Figure 2: Visualization of pixel-level forensic signals extracted from an input image. From left to right: the original image, the Laplacian response highlighting edge sharpness and local inconsistencies, the LBP texture map capturing micro-texture patterns associated with smoothing or synthesis, and the FFT magnitude spectrum revealing frequency-domain artifacts indicative of generative or post-processing operations.

Module 2: Semantic Reasoning via VLMs

Module 2 performs high-level semantic reasoning over the outputs of Module 1 using a Vision-Language Model, specifically **Qwen2-2B**. The VLM integrates the classifier probabilities, extracted forensic signals, and visual content to assess the plausibility of the predicted manipulation type. By reasoning about semantic and physical consistency, the model generates an interpretable explanation and produces a structured output containing the final manipulation type, authenticity confidence, and reasoning.

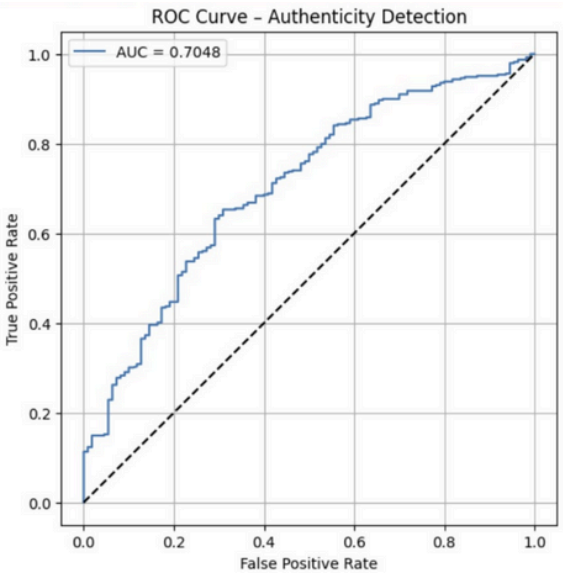
{'predicted_label': 'Deepfake',
'confidence': 0.933851957321167},
{'texture_laplacian': 608.8340815278405,
'lbp_texture': 6.005541142017244,
'fft_frequency': 8.531407866687777,
'dct_compression': 15.761184692382812}

VLM
Qwen2-2B

" The face appears unusually smooth and does not blend naturally with the surrounding scene. These visual inconsistencies suggest the image has been digitally manipulated rather than captured as-is."

EVALUATION

By merging the Artificial and Deepfake classes into a single manipulated category and evaluating against real images, the model achieves an **AUC of 0.70**, indicating moderate discriminative capability. This improvement suggests that while the model struggles to distinguish between different manipulation types, it is more effective at identifying manipulated content in general. However, the remaining overlap between manipulated and real samples highlights the difficulty of detecting subtle or high-quality synthetic images and motivates the need for additional semantic reasoning and forensic fusion.



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